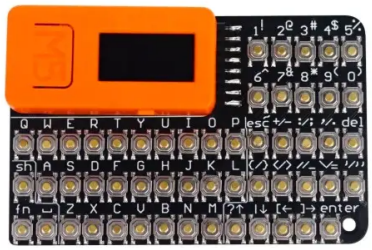
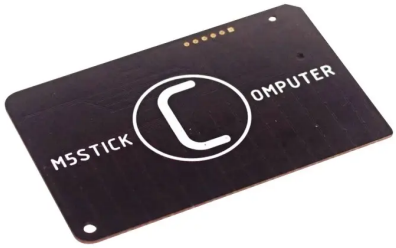
Hat CardKB

SKU:U077







Description

Hat CardKB is a fully functional QWERTY keyboard. If you want to achieve some complex keyboard input interactions, relying solely on the buttons on the M5StickC can be quite challenging. To solve this problem, we have introduced the **Hat CardKB**.

Using the **Hat CardKB** not only allows for full keyboard input but also supports various key combinations (Shift + Key, Fn + Key) to output richer key values. An onboard colorful LED can provide different color light prompts based on the input mode. The I2C address of **Hat CardKB** is 0x5F.

Features

- Full keyboard input with various key combinations.
- Compact and portable.

Includes

- 1 x Hat CardKB

Applications

- M5StickC keyboard input peripheral

Specifications

Specification	Parameter
Communication Port	I2C @0x5F
Product Size	84.6 x 54.2 x 6.5mm
Product Weight	17.0g
Package Size	115.0 x 96.0 x 40.0mm
Gross Weight	21.0g

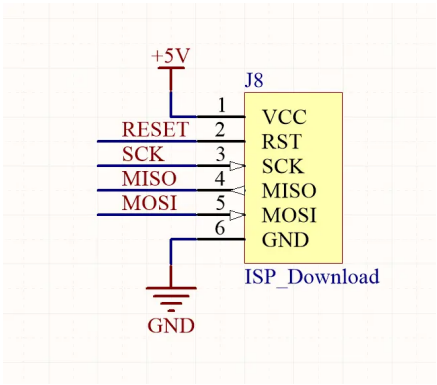
Button Combination Instructions:

- **Press a single key** (blue light flashes once), the keyboard will output the first key value (for letter keys, it outputs in lowercase). For example, pressing "Q" will output "q" (lowercase).
- **Double-click Shift or Fn**, to lock Shift (red light stays on) or Fn (green light stays on), making it convenient to output the second or third key value multiple times.
- **Shift+key** (red light flashes), the keyboard will output the uppercase form of the letter, and reuse keys will output the second key value. For example, after clicking "Shift" once, pressing "Q" will output "Q" (uppercase). Double-clicking "Shift" locks the function, and any letter key pressed afterward will output in uppercase, while number and symbol keys will output the second key value. Click "Shift" again to unlock.
- **Fn+key (custom function key combination)** (green light flashes), the keyboard will output the third key value. You can customize the function corresponding to the pressed key.

symbol											1 !	2 @	3 #	4 $	5 %
Singal Key											0x31	0x32	0x33	0x34	0x35
Shift+Key											0x21	0x40	0x23	0x24	0x25
Fn+Key											0x80	0x81	0x82	0x83	0x84
symbol											6 ^	7 &	8 *	9 (	0)
Singal Key											0x36	0x37	0x38	0x39	0x30
Shift+Key											0x5E	0x26	0x2A	0x28	0x29
Fn+Key											0x85	0x86	0x87	0x88	0x89
symbol	q Q	w W	e E	r R	t T	y Y	u U	i I	o O	p P	esc ~	+ =	: ;	, .	del
Singal Key	0x71	0x77	0x65	0x72	0x74	0x79	0x75	0x69	0x6F	0x70	0x1B	0x2B	0x3A	0x2C	0x08
Shift+Key	0x51	0x57	0x45	0x52	0x54	0x59	0x55	0x49	0x4F	0x50	0x7E	0x2D	0X3B	0X2E	
Fn+Key	0x8A	0x8B	0x8C	0x8D	0x8E	0x8F	0x90	0x91	0x92	0x93	0x94	0x95	0x96	0x97	0x98
symbol	sh	a A	s S	d D	f F	g G	h H	j J	k K	l L	{ }	< >	/ _	\ =	' "
Singal Key	0x80	0x61	0x73	0x64	0x66	0x67	0x68	0x6A	0x6B	0x6C	0x7B	0x3C	0x2F	0x5C	0x27
Shift+Key		0x41	0x53	0x44	0x46	0x47	0x48	0x4A	0x4B	0x4C	0x7D	0x3E	0x5F	0x3D	0x22
Fn+Key		0x99	0x9A	0x9B	0x9C	0x9D	0x9E	0x9F	0xA0	0xA1	0xA2	0xA3	0xA4	0xA5	0xA6
symbol	fn	space	z Z	x X	c C	v V	b B	n N	m M	? ↑	| ↓	[←	] →	enter	
Singal Key	0x81	0x20	0x7A	0x78	0x63	0x76	0x62	0x6E	0x6D	0x3F	0x7C	0x5B	0x5D	0x0D	
Shift+Key			0x5A	0x58	0x43	0x56	0x42	0x4E	0x4D	0x99	0xA4	0x98	0xA5		
Fn+Key		0xA7	0xA8	0xA9	0xAA	0xAB	0xAC	0xAD	0xAE	0xAF	0xB0	0xB1	0xB2	0xB3	

PinMap

Mega328 ISP download interface pin definition



Softwares

Arduino

- Hat CardKB Example

UiFlow1

- Hat CardKB UiFlow1 Docs

UiFlow2

- Hat CardKB UiFlow2 Docs

Internal Firmware

- Hat CardKB Internal Firmware

Protocol

- Protocol Type: I2C
- I2C Address: **0x5F**

```
/*-------------------------------------------------------------------------------------------------------*/
| KEYBOARD REG       | 0x5F
| ------------------------------------------------------------------------------------------------------
| keyboard_value_reg[0]        |  R |  KEYBOARD VALUE
/*-------------------------------------------------------------------------------------------------------*/
```

EasyLoader

Easyloader	Download Link	Remarks
Hat CardKB Easyloader	download	/

Video

Usage demonstration of Hat CardKB

CardKB_HAT.mp4